Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 12 GB of memory. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the validation accuracy by dataset.

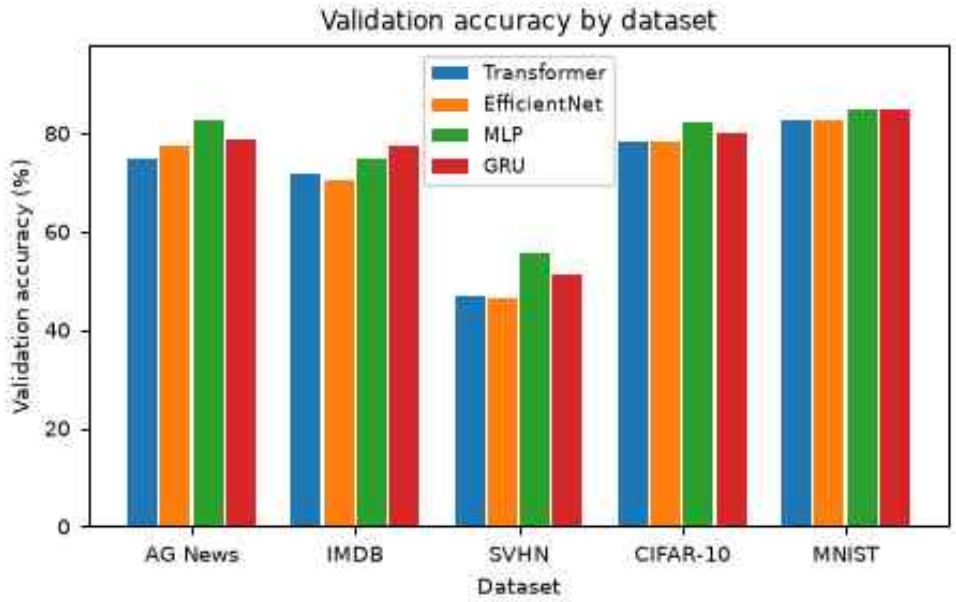


Figure 1: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 1, Transformer reaches 25.7% validation accuracy on AG News. On IMDB, MLP is 4.2 points better than EfficientNet. MLP achieves over 80% validation accuracy on CIFAR-10 (Figure 1).

Figure 2 shows the f1 score during training.

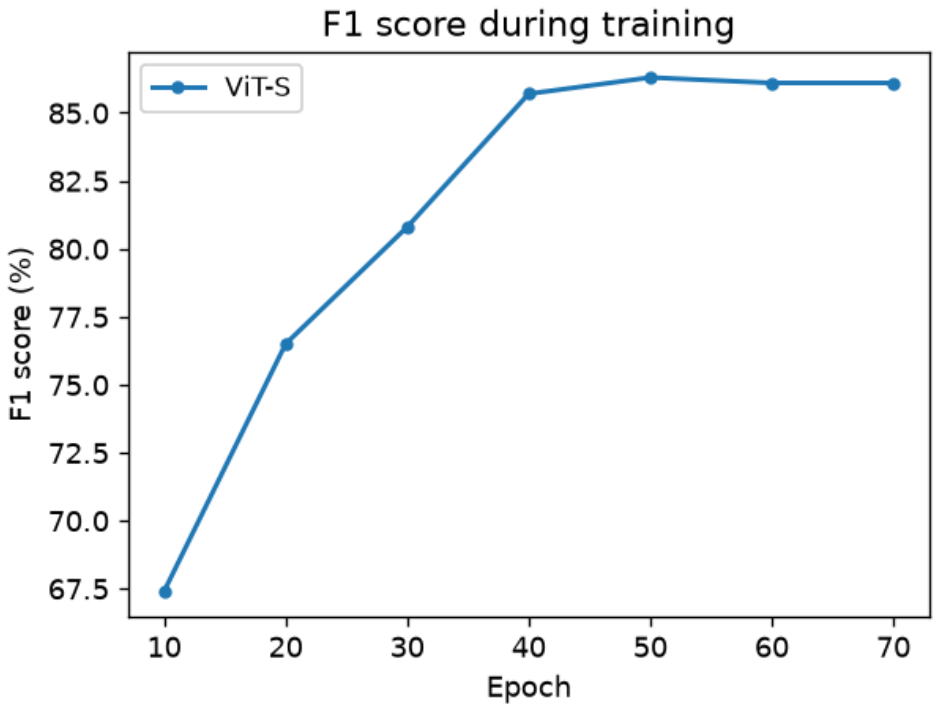


Figure 2: F1 score during training for the models in this experiment.

ViT-S scores 10 points higher on 70 than on 20. As shown in Figure 2, ViT-S scores 18.9 points higher on 50 than on 10. As shown in Figure 2, ViT-S reaches 86.1% F1 score at epoch 70.

Figure 3 shows the accuracy by dataset.

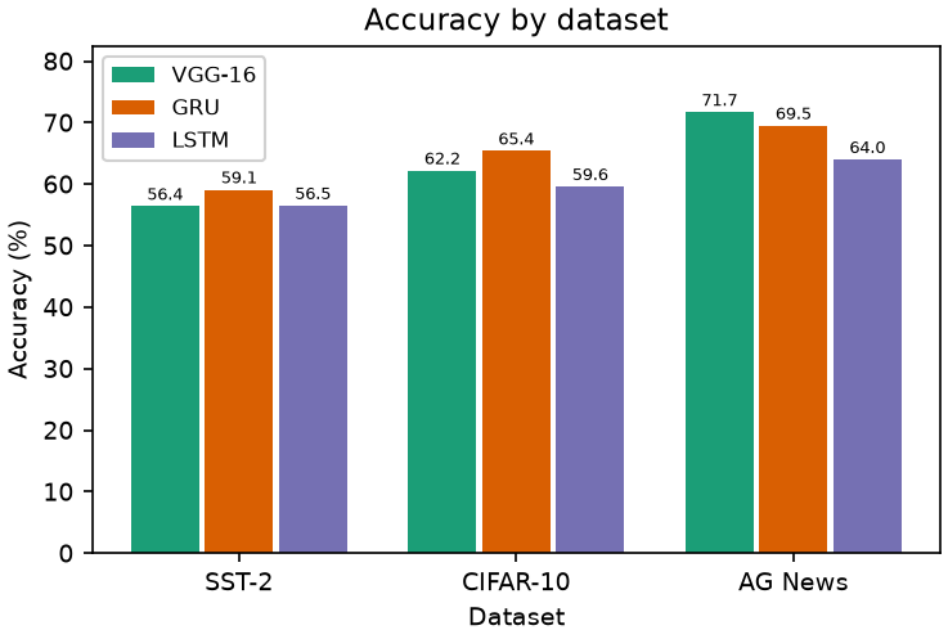


Figure 3: Accuracy by dataset for the models in this experiment.

LSTM achieves 51.4% accuracy on CIFAR-10. As shown in Figure 3, LSTM achieves 59.6% accuracy on CIFAR-10.

Figure 4 shows the test accuracy of ResNet-18 by dataset.

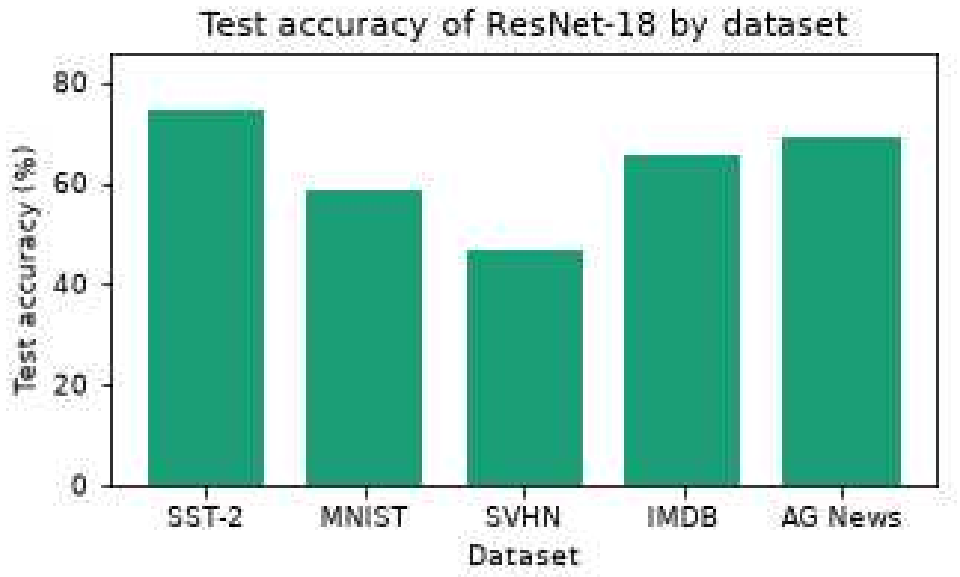


Figure 4: Test accuracy of ResNet-18 by dataset for the models in this experiment.

ResNet-18 reaches over 70% test accuracy on SST-2 (Figure 4). As shown in Figure 4, ResNet-18 scores 54.6 points higher on MNIST than on SVHN. As shown in Figure 4, ResNet-18 reaches 66% test accuracy on IMDB.

3 Discussion

On CIFAR-10, XGBoost obtains an accuracy of 68.9%. The differences between the models are discussed in the next section.